

Alessio Salvatore

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Education

ETH Zürich, Zürich, Switzerland Sep 2025–Present
M.Sc. in Robotics
Coursework: Probabilistic Artificial Intelligence, Advanced Machine Learning, Image Analysis and Computer Vision, Planning and Decision Making for Autonomous Robots, 3D Vision, Perception and Learning for Robotics, Deep Generative Models, Robot Learning

Politecnico di Milano, Milan, Italy Sep 2022–Jul 2025
B.Sc. in Automation Engineering
Final Grade: 110/110 *cum laude* (equivalent GPA $\approx 4.0/4.0$; 29.57/30).

Illinois Institute of Technology, Chicago, IL, USA Aug–Dec 2024
Exchange — Global E3 Program

Projects

Current Projects

Task-Aware 3D Map Generation with RL Feb 2026–Present
Computer Vision and Geometry Group – Marc Pollefeys Lab

- Training an RL agent to dynamically select voxel resolution in 3D maps based on task relevance, enabling efficient scene representations for downstream navigation
- Integrating a 3D reconstruction pipeline into the PEANUT codebase to support point-goal and object-goal navigation tasks

Knowledge Distillation for Aerial Robots Control Feb 2026–Present
Autonomous system Lab – Roland Siegwart Lab

- Selected among 150 applicants for a closed cohort of 30 students in the Perception and Learning class at ETH Zurich
- Developing an imitation learning pipeline for OMAV (Omnidirectional Micro Aerial Vehicle) control, exploring Behavioral Cloning and DAgger approaches
- Designing and benchmarking neural approximations of a model-based controller
- Deploying the learned policy on hardware, leveraging real-flight data augmentation to bridge the sim-to-real gap

Past Projects

DINOv3 for Mitral Valve Segmentation  Repo 12/2025

- Built a unified preprocessing pipeline for heterogeneous medical video datasets
- Leveraged DINO-based feature propagation to generate pseudo-labels from sparse annotations
- Trained a U-Net segmentation model using filtered pseudo-labels

Multi-Agent Multi-Goal Collection Planner 12/2025
Project of the Class Planning and Decision Making for Autonomous Robots

- Designed a multi-robot planning framework combining Large Neighborhood Search for task allocation with real-time path planning and decentralized collision avoidance under partial observability.

Autonomous Satellite Docking Planner 11/2025 – 12/2025
Project of the Class Planning and Decision Making for Autonomous Robots

- Implemented a trajectory optimization pipeline for autonomous satellite docking, solving non-convex guidance problems via Sequential Convexification in cluttered environments.

Work Experience

JOINT Lab — Italian Institute of Technology, Bergamo Feb–Jun 2025
Research Intern — Industrial Robotics  Repo

- CAD design of a linear actuator built from commercial off-the-shelf components.
- Implemented real-time control algorithms using Arduino.
- Developed a MATLAB simulation to validate system dynamics and behavior.

ETH Robotics Club Nov 2025–Present
Partnerships Team Lead

- Secured sponsorships for club initiatives and events